TÉCNICO
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SHIP AND OCEAN SYSTEMS DESIGN

Ship types

Twelve groups, and what sets the size of each

Ship and Ocean Systems Design (PNSO)
Academic year 2026/2027

In this lecture

1. A classification of ships in twelve groups, from dry cargo to offshore wind.
2. For each group, the requirement that sets the main dimensions: deadweight, volume, deck area, speed or mission.
3. Where the displacement equation $\Delta = DWT + lightship$ decides the size, and where it is only a check.

TABLE 1 · CLASSIFICATION (1 OF 2)

Main ship types

1 · Dry cargo

General cargo and multi-purpose ship · Container ship · Ro-Ro and Con-Ro · Car carrier (PCC, PTCC) · Heavy-lift and semi-submersible ship · Reefer · Livestock carrier · Bulk carrier (Handysize to Capesize) · Ore carrier · Self-unloader

2 · Liquid cargo

Crude oil tanker (Aframax, Suezmax, VLCC) · Product tanker · Chemical tanker · Shuttle tanker · Bunker tanker · LPG carrier · LNG carrier · Ammonia and CO2 carrier

3 · Passenger ships

Cruise ship · Expedition cruise ship · River cruise vessel · Ro-Pax ferry · Car and passenger ferry · Passenger ferry · Excursion vessel

4 · Naval vessels

Aircraft carrier · Amphibious transport vessel · Logistic support vessel · Frigate · Corvette · Patrol vessel · Mine countermeasure vessel · Landing craft · Submarine · Auxiliary support vessel · Uncrewed surface vessel (USV)

5 · Fishing vessels

Stern trawler · Freezer and factory trawler · Purse seiner · Longliner · Gillnetter · Pole-and-line vessel · Fish carrier

6 · Dredging vessels

Trailing suction hopper dredger (TSHD) · Cutter suction dredger (CSD) · Backhoe dredger · Grab dredger · Bucket ladder dredger · Split-hopper barge · Rock dumper and fall-pipe vessel · Water injection dredger

Merchant ships: 1–3 · Government, fishing and work: 4–8 · Leisure and fast craft: 9–10 · Offshore energy: 11–12

TABLE 1 · CLASSIFICATION (2 OF 2)

Main ship types

7 · Work ships

Crane vessel · Shear-leg crane · Buoy tender · Oil-recovery vessel

8 · Auxiliary vessels

Seagoing tug · Harbour tug (ASD, tractor) · Escort tug · Icebreaker · Pilot vessel · Coastguard and SAR vessel · Fire-fighting vessel · Research vessel · Hydrographic survey vessel · Training ship

9 · Yachts

Motor yacht · Superyacht (over 24 m) · Explorer yacht · Sailing yacht · Yacht support vessel

10 · Fast craft

Planing monohull · Catamaran and wave-piercer · Axe bow · SWATH · Hydrofoil · Surface effect ship (SES) · Hovercraft (ACV) · Crew transfer vessel (CTV)

11 · Offshore oil and gas

Semi-submersible drilling rig · Jack-up · Drillship · FPSO and FSO · FLNG · Spar · Tension-leg platform (TLP) · Seismic survey vessel · Platform supply vessel (PSV) · Anchor handling tug supply (AHTS) · Diving support vessel (DSV) · Construction support vessel (CSV) · Multipurpose support vessel · Pipe-layer · Well service vessel

12 · Offshore wind

Wind turbine installation vessel (WTIV) · Cable-layer · Service operation vessel (SOV) · Construction support vessel (CSV) · Accommodation unit (flotel) · Bottom-fixed and floating wind turbines

Merchant ships: 1–3 · Government, fishing and work: 4–8 · Leisure and fast craft: 9–10 · Offshore energy: 11–12

TABLE 2 · WHAT SETS THE SIZE

Five design drivers across the twelve groups

Size set by	Typical ships (groups in Table 1)	Role of Δ = DWT + lightship
Deadweight	Bulk and ore carriers, crude and product tankers, hopper dredgers (1, 2, 6)	**Sizes the ship:** Δ follows from DWT, then L, B, T
Cargo volume	Container, Ro-Ro, PCC/PTCC, reefer, livestock, LPG/LNG carriers (1, 2)	Volume fixes L, B, D; the equation gives the draught
People and spaces	Passenger ships, ferries, cruise ships, yachts (3, 9)	Decks, gross tonnage and stability govern; weight is a check
Deck area and lift	Crane and supply vessels, AHTS, heavy-lift ships, offshore units, F(P)SO, wind installation vessels (1, 7, 11, 12)	Deck area, deck load and stability govern; Δ checks the payload
Power, speed or mission	Tugs, icebreakers, fishing, naval vessels, fast craft (4, 5, 8, 10)	**Weight-critical:** lightship grows with power and speed

The exercise on the displacement equation is the first row: a deadweight carrier, where solving Δ = DWT + lightship(Δ) gives the size of the ship.